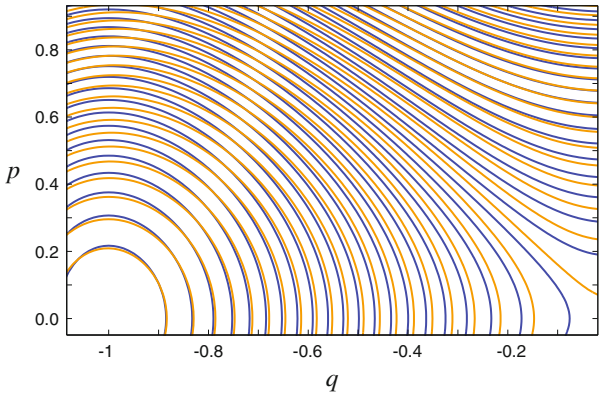


Fig. 3.3 The isoenergy contours are shown for the double well model (*dark curves*) together with the corresponding curves for the Verlet modified Hamiltonian (*light curves*)

The implication of backward error analysis is that the convergence order is still directly relevant in molecular simulations, even when the accuracy of trajectories (described by the convergence theorem) cannot be verified. We can think of the modified energy surface as a rippled version of the original. The order determining the allowed magnitude of the fluctuation.

To illustrate this idea, we have plotted, in Fig. 3.3 the contours of the energy and the Verlet modified energy (including terms through h^4) for the double well example (potential $U(q) = (q^2 - 1)^2/2$) in part of phase space. The figure shows that the contours of the two energy functions are interleaved. If we are confined to the contours of the modified energy the physical energy will be seen to fluctuate along it.[1]

Suppose that, somehow, H were exactly conserved along the numerical solution, so

$$\dot{H} = 0 \Rightarrow \{H, \tilde{H}_h\} = 0.$$

[1]The modified energy for the Verlet method for a single degree of freedom system with energy $H = p^2/2 + U(q)$ is

$$\begin{aligned}\tilde{H}_h = H &+ \frac{h^2}{24}(2p^T U'' p - (U')^2) \\ &+ h^4\left(\frac{1}{720}p^4 U'''' - \frac{1}{120}p^2 U' U''' - \frac{1}{240}(U')^2 U'' - \frac{1}{60}p^2((U'')^2 + U' U''')\right) \\ &+ \mathcal{O}(h^6)\end{aligned}$$